

The intermediate singular-continuous spectrum is unavoidable

A full answer to item (c) of Theorem 2.6 in arXiv:1106.4852

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Status. Candidate full answer to the explicit alternative in item (c), likely valid, pending human review. Under the one-dimensional spectral conclusions used by the source, the singular-continuous spectrum of its two-dimensional separable operator is not merely nonempty in the intermediate band: it contains that entire band. The other open problems in the source are not addressed.

1 Source question

Marchetti and Wreszinski consider two independent copies J_1, J_2 of a random sparse Jacobi operator and the Kronecker sum

$$K_\theta = J_1 \otimes I + \theta I \otimes J_2, \quad 0 < \theta \leq 1.$$

Write $L = \lambda^+ = -\lambda^-$, where $0 < L < 2$. Item (c) of their Theorem 2.6 leaves open whether

$$\sigma_{\text{sc}}(K_\theta) \cap (-L(1 + \theta), L(1 + \theta))$$

is empty [1].

b.

$$[-2(1 + \theta), \lambda^-(1 + \theta)) \cup (\lambda^+(1 + \theta), 2(1 + \theta)] \subset \sigma_{\text{pp}}(J_\theta^{(2)}) \quad (2.20c)$$

c.

$$\sigma_{\text{sc}}(J_\theta^{(2)}) \cap (\lambda^-(1 + \theta), \lambda^+(1 + \theta)) \quad (2.20d)$$

may, or may not, be an empty set.

The conclusion explicitly asks for clarification of this alternative.

Finally, it is clear that, besides the intermediate region mentioned above, Theorem 2.6 leaves much room for improvement. Elimination of the set of zero Lebesgue measure in the s.c. part of the spectrum would be a significant improvement, as well as clarification of which alternative holds in item c. of Theorem 2.6

The one-dimensional theorem on which the source relies supplies, for the generic parameters and configurations in question,

$$[-L, L] \subseteq \sigma(J_i | \mathcal{H}_{\text{sc}}^{(i)}), \quad [-2, -L] \cup [L, 2] \subseteq \sigma(J_i | \mathcal{H}_{\text{pp}}^{(i)}), \quad i = 1, 2. \quad (1)$$

Indeed, the exceptional set in the singular-continuous region has Lebesgue measure zero and hence dense complement, while the pure-point spectrum is dense in the two outer intervals. Taking closures gives (1).

2 Proof intuition

The mixed tensor sectors were not exploited in item (c). If u is an eigenvector of J_1 with eigenvalue e , then $\mathbb{C}u \otimes \mathcal{H}_{\text{sc}}^{(2)}$ is a reducing subspace for K_θ , and the restriction is unitarily equivalent to $e + \theta J_2|_{\mathcal{H}_{\text{sc}}^{(2)}}$. It is therefore singular continuous. Summing over the countably many eigenvectors preserves singular continuity: every scalar spectral measure is a countable sum of translated singular continuous measures.

Consequently the singular-continuous spectrum contains both mixed Minkowski sums

$$\sigma(J_1|_{\mathcal{H}_{\text{pp}}^{(1)}}) + \theta\sigma(J_2|_{\mathcal{H}_{\text{sc}}^{(2)}}) \quad \text{and} \quad \sigma(J_1|_{\mathcal{H}_{\text{sc}}^{(1)}}) + \theta\sigma(J_2|_{\mathcal{H}_{\text{pp}}^{(2)}}).$$

For the inner interval $[-L, L]$ and outer intervals $[-2, -L] \cup [L, 2]$, these two sums overlap and fill one large interval.

3 The mixed-sector lemma

Lemma 1. *Let A and B be bounded self-adjoint operators on separable Hilbert spaces, and let $\theta > 0$. For*

$$T = A \otimes I + \theta I \otimes B,$$

the reducing subspaces

$$\mathcal{H}_{\text{pp}}(A) \otimes \mathcal{H}_{\text{sc}}(B), \quad \mathcal{H}_{\text{sc}}(A) \otimes \mathcal{H}_{\text{pp}}(B)$$

are purely singular continuous. In particular,

$$\begin{aligned} \sigma_{\text{sc}}(T) \supseteq & (\sigma(A|_{\mathcal{H}_{\text{pp}}(A)}) + \theta\sigma(B|_{\mathcal{H}_{\text{sc}}(B)})) \\ & \cup (\sigma(A|_{\mathcal{H}_{\text{sc}}(A)}) + \theta\sigma(B|_{\mathcal{H}_{\text{pp}}(B)})). \end{aligned}$$

Proof. Choose an orthonormal eigenbasis (u_n) for $\mathcal{H}_{\text{pp}}(A)$, repeating eigenvalues according to multiplicity, and write $Au_n = a_n u_n$. Under the canonical identification

$$\mathcal{H}_{\text{pp}}(A) \otimes \mathcal{H}_{\text{sc}}(B) \simeq \bigoplus_n \mathcal{H}_{\text{sc}}(B),$$

the restriction of T is

$$\bigoplus_n (a_n + \theta B|_{\mathcal{H}_{\text{sc}}(B)}).$$

For a vector (x_n) , its scalar spectral measure is the countable sum of the translated and rescaled singular-continuous measures of the x_n . This sum has no atoms. It is singular because each summand is carried by a Lebesgue-null set and a countable union of null sets is null. Hence the restriction is purely singular continuous.

Its spectrum is the closure of $\bigcup_n (a_n + \theta\sigma(B|_{\mathcal{H}_{\text{sc}}(B)}))$. Since the eigenvalues a_n are dense in the spectrum of the pure-point restriction and all spectra here are compact, this closure equals

$$\sigma(A|_{\mathcal{H}_{\text{pp}}(A)}) + \theta\sigma(B|_{\mathcal{H}_{\text{sc}}(B)}).$$

The second mixed sector follows by interchanging A and B . □

4 Resolution of item (c)

Theorem 2. *Under the hypotheses and almost-everywhere one-dimensional spectral conclusions used in Theorem 2.6 of the source, for every $0 < \theta \leq 1$,*

$$[-2 - \theta L, 2 + \theta L] \subseteq \sigma_{\text{sc}}(K_\theta). \tag{2}$$

In particular,

$$(-L(1 + \theta), L(1 + \theta)) \subseteq \sigma_{\text{sc}}(K_\theta), \quad (3)$$

so the set in item (c) is never empty in this regime.

Proof. Put

$$I = [-L, L], \quad P = [-2, -L] \cup [L, 2].$$

Equations (1) and Lemma 1 give

$$\sigma_{\text{sc}}(K_\theta) \supseteq (P + \theta I) \cup (I + \theta P).$$

The four component intervals are

$$\begin{aligned} P + \theta I &= [-2 - \theta L, -L(1 - \theta)] \cup [L(1 - \theta), 2 + \theta L], \\ I + \theta P &= [-L - 2\theta, L(1 - \theta)] \cup [-L(1 - \theta), L + 2\theta]. \end{aligned}$$

Because $0 < \theta \leq 1$ and $0 < L < 2$, consecutive intervals overlap, the leftmost endpoint is $-2 - \theta L$, and the rightmost endpoint is $2 + \theta L$. Their union is exactly the interval in (2).

Finally $L \leq 2$ implies $L(1 + \theta) \leq 2 + \theta L$. The symmetric inequality at the left endpoint proves (3). $\square$

5 Strength, extensions, and novelty

The first attack used one pure-point eigenvalue close to L and one singular-continuous spectral point close to $-L$, which already proves nonemptiness. The deeper upgrade above uses both mixed reducing sectors and their full spectra, yielding the interval inclusion (2). This also shows that the source’s pure-point and singular-continuous spectral sets can overlap topologically; no contradiction arises because spectral-type subspaces, rather than their spectra as closed sets, are mutually orthogonal.

The same argument extends to the source’s separable d -dimensional variants: any tensor sector with exactly one singular-continuous factor and all remaining factors pure point is purely singular continuous, and its spectrum is the corresponding Minkowski sum. This observation does not prove a mobility edge or pointwise Fourier–Stieltjes decay, nor does it remove the one-dimensional exceptional null set; those separate source questions remain open.

A bounded primary-source novelty search covered the exact title and authors, the phrases “singular continuous” and “Kronecker sum”, the stated item (c) alternative, and the authors’ related arXiv papers on sparse Jacobi matrices. It found the source and its one-dimensional predecessors, but no later primary paper giving this mixed-sector answer. Novelty confidence is moderate because the argument is elementary; correctness confidence is high. Human review should verify that the source’s one-dimensional type statements indeed give the closed spectral inclusions (1) in precisely the boundary-phase regime intended by Theorem 2.6.

References

- [1] D. H. U. Marchetti and W. F. Wreszinski, *Anderson-like transition for a class of random sparse models in $d \geq 2$ dimensions*, arXiv:1106.4852.